# Nios® V/m Processor Helloworld and OCM test design

## Agilex™ 5 FPGA

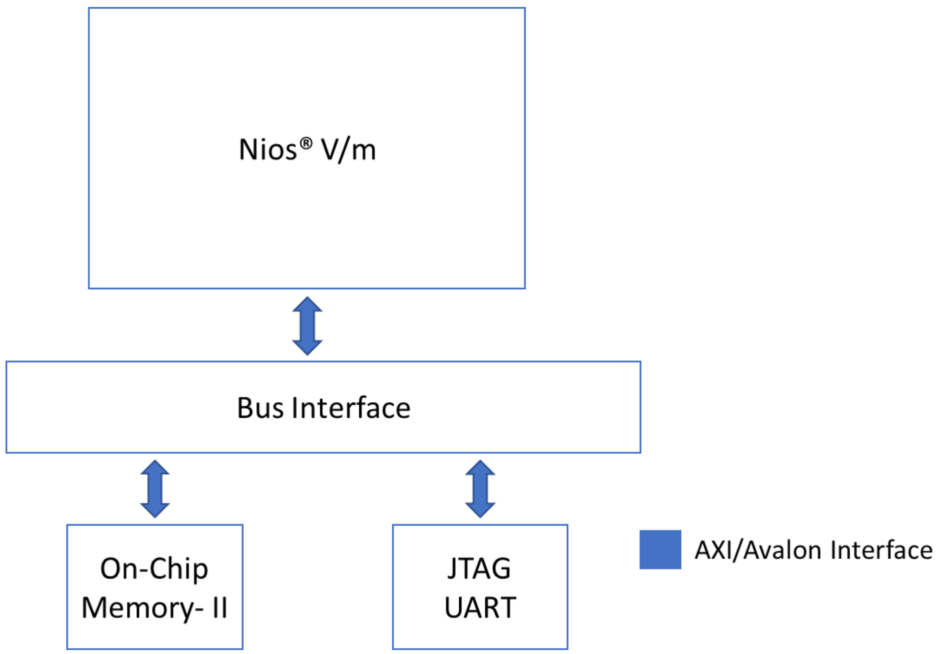
Last updated: **2024-09-30**

# Contents

# 1.0 Theory of Operation

Nios® V/m Processor-based Helloworld example design on the Agilex™ 5 FPGA E-Series 065B Premium Development Kit (ES1) DK-A5E065BB32AES1.

**Figure 1-1. Block Diagram**




## 1.1 IP Cores

The following IPs are used in this design.

- Nios V/m soft processor core
- On Chip RAM-II
- JTAG UART
- System ID

# 2.0 Simulating the design

*Note*: Please refer to the README.md file in the extracted design package from Quartus or in the sources folder of the Github repository of the design for the steps to create the design, application and generate the programing files.

- Unpackage/extract the design in your working directory.
- Simulate the design using Quartus Prime Pro tool.

# 3.0 Executing the design on Development Kit

## 3.1 Creating and validating the design

*Note*: Please refer to the README.md file in the extracted design package from Quartus or in the sources folder of the Github repository of the design for the steps to create the design, application and generate the programing files.

- Unpackage/extract the design in your working directory.
  Locate the "ready_to_test" folder within the package.
- The folder contains the necessary files for executing the application on the board.
  Refer to the readme file for the steps to program the application files on the board.
- Validate the design by observing the prints on the terminal.

# 4.0 Expected Results

The following is the output as observed on the JTAG UART terminal. The output is analogous to the logic from the application code. Users should be able to observe same output on their terminal/setup.

```
# Hello World from SM:Agilex-5  NIOSV/m core!
# Application will execute Memory Test
# Starting Memory test
# value at memory location 0x0 with offset 0 is 0x4080006f
# After write: value at memory location 0x0 with offset 0 is 0xa5a5a5a5
# Memory write test PASSED
# value at memory location 0x0 with offset 4 is 0x0
# After write: value at memory location 0x0 with offset 4 is 0xa5a5a5a5
# Memory write test PASSED
# value at memory location 0x0 with offset 8 is 0x0
# After write: value at memory location 0x0 with offset 8 is 0xa5a5a5a5
# Memory write test PASSED
# value at memory location 0x0 with offset 12 is 0x0
# After write: value at memory location 0x0 with offset 12 is 0xa5a5a5a5
# Memory write test PASSED
# value at memory location 0x0 with offset 16 is 0x0
# After write: value at memory location 0x0 with offset 16 is 0xa5a5a5a5
# Memory write test PASSED
# value at memory location 0x0 with offset 20 is 0x0
# After write: value at memory location 0x0 with offset 20 is 0xa5a5a5a5
# Memory write test PASSED
# value at memory location 0x0 with offset 24 is 0x0
# After write: value at memory location 0x0 with offset 24 is 0xa5a5a5a5
# Memory write test PASSED
# value at memory location 0x0 with offset 28 is 0x0
# After write: value at memory location 0x0 with offset 28 is 0xa5a5a5a5
# Memory write test PASSED
# Memory Test Complete
# Print the value of System ID
# System ID from Peripheral core is 0xA5
#
```

# 5.0 Document Revision History

| Date | Version | Changes |
|---|---|---|
| 2024-09-30 | 24.3.0 | • Initial release on github |